

Name _____

Exam III-B
Make-up

Score _____

Section I. Short Answer Problems. Answer the problems on separate paper. You do not need to rewrite the problem statements on your answer sheets. Work carefully. Do your own work. **Show all relevant supporting steps!** *STAPLE* this sheet to the front of your answers.

1. (4 pts) Consider the vector space $\mathbb{R}^4$ with basis $B = \{(-1, 0, 2, 1)^T, (2, -1, 3, 0)^T, (0, 2, -1, -1)^T, (3, -1, 0, 2)^T\}$. Suppose that $[x]_B = (-2, 4, 1, -3)^T$. Find the standard coordinates of x .

2. (8 pts) Consider the vector space $\mathbb{R}^4$ with basis $B = \{(-1, 0, 1, -1)^T, (1, -1, 0, 1)^T, (-1, 1, -1, 0)^T, (0, 1, -1, 1)^T\}$. Suppose that $x = (-2, 4, 1, -3)^T$. Find $[x]_B$.

3. (10.5 pts) Consider the vector space $\mathbb{R}^5$. Consider the vectors $u = (-1, 2, 0, 2, 3)^T$, $v = (0, -2, 1, -1, 2)^T$, $w = (-1, 1, -1, 3, 2)^T$. Find

$$A. \|u\|, B. \|v + w\|, C. d(u, v), D. \cos \theta,$$

where θ is the angle between v and w . Also, determine whether any of the following pairs of vectors are orthogonal, parallel or neither:

- a. u and v , b. v and w , c. $u - v$ and w

4. (10.5 pts) Consider the inner product space $\mathbb{R}^5$ with inner product $\langle u, v \rangle = u_1v_1 + 3u_2v_2 + 4u_3v_3 + 2u_4v_4 + 4u_5v_5$ for $u = (u_1, u_2, u_3, u_4, u_5)^T$ and $v = (v_1, v_2, v_3, v_4, v_5)^T$. Consider the vectors $u = (-1, 2, 0, 2, 3)^T$, $v = (0, -2, 1, -1, 2)^T$, $w = (-1, 1, -1, 3, 2)^T$. Find

$$A. \|u\|, B. \|v + w\|, C. d(u, v), D. \cos \theta,$$

where θ is the angle between v and w . Also, determine whether any of the following pairs of vectors are orthogonal, parallel or neither:

- a. u and v , b. v and w , c. $u - v$ and w

5. (10.5 pts) Consider the inner product space P_2 with (coefficient) inner product $\langle p, q \rangle = a_0 b_0 + a_1 b_1 + 2a_2 b_2$ for $p(x) = a_0 + a_1 x + a_2 x^2$ and $q(x) = b_0 + b_1 x + b_2 x^2$. Consider the vectors $u = 1 + 2x - 2x^2$, $v = 2 - x^2$, $w = -1 + x - x^2$. Find

$$A. \|u\|, B. \|v + w\|, C. d(u, v), D. \cos \theta,$$

where θ is the angle between v and w . Also, determine whether any of the following pairs of vectors are orthogonal, parallel or neither:

- a. u and v , b. v and w , c. $u - v$ and w

6. (8 pts) Consider the inner product space $\mathbb{R}^5$ with inner product $\langle u, v \rangle = u_1 v_1 + 3u_2 v_2 + 4u_3 v_3 + 2u_4 v_4 + 4u_5 v_5$ for $u = (u_1, u_2, u_3, u_4, u_5)^T$ and $v = (v_1, v_2, v_3, v_4, v_5)^T$. Consider the vectors $u = (-1, 2, 0, 2, 3)^T$, $v = (0, -2, 1, -1, 2)^T$, $w = (-1, 1, -1, 3, 2)^T$. Find $\text{proj}_u v$ and $\text{proj}_v w$.

7. (10 pts) Consider the inner product space $\mathbb{R}^3$ with standard inner product (dot product). Let S be the subspace of $\mathbb{R}^3$ which is the span of the vectors $u = (1, 1, 0)^T$ and $v = (1, 0, -1)^T$. Find the projection of $w = (2, 3, 4)^T$ on to S .